

Decentralization

Outline



- Decentralization using blockchain
- Methods of decentralization
- Routes to decentralization
- Blockchain and full ecosystem decentralization
- Platforms for decentralization

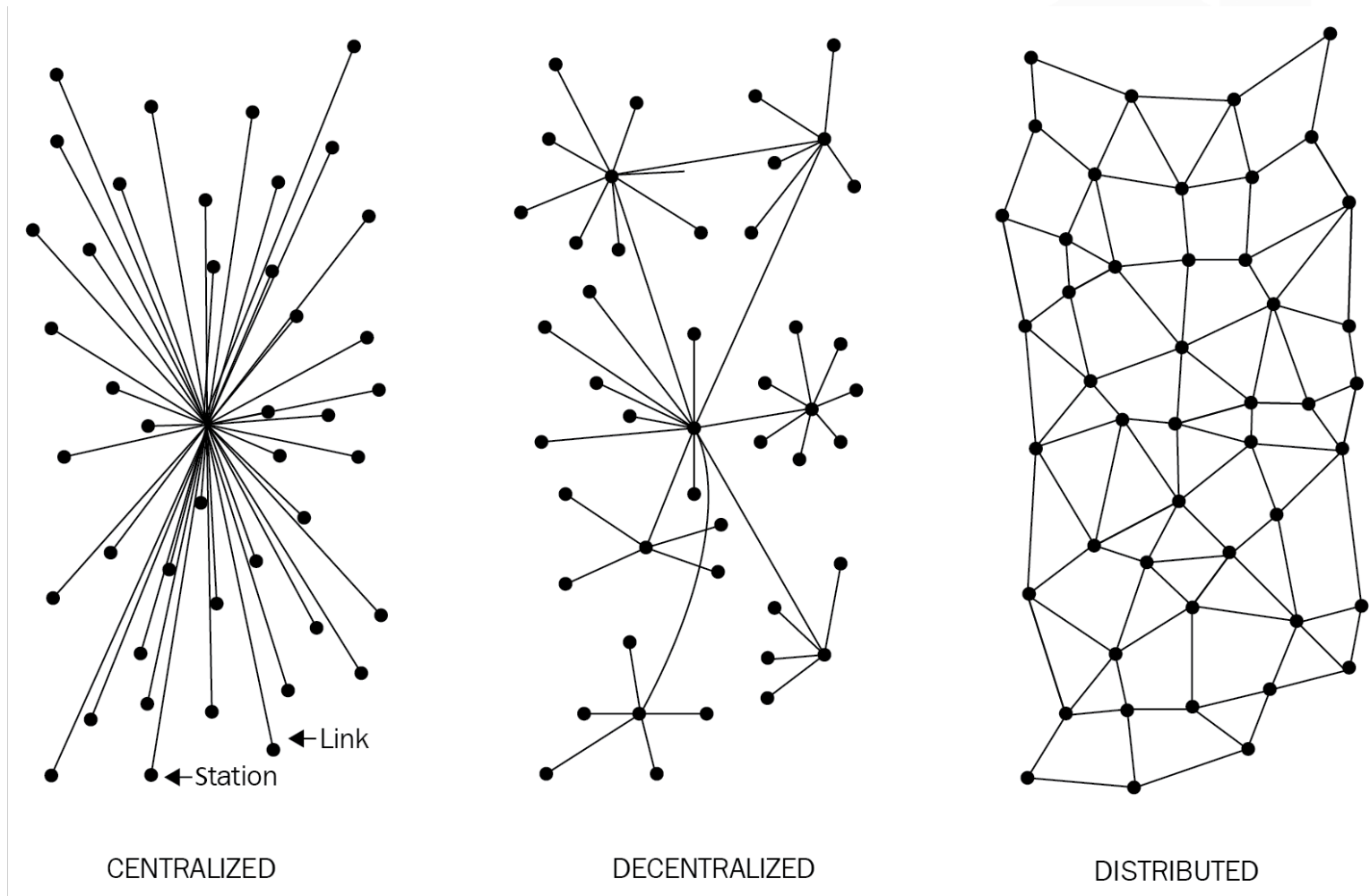
Decentralization using blockchain



- Decentralization is a core benefit and service provided by blockchain technology.
- Provide a platform that does not need any intermediaries and can function with many different leaders chosen via consensus mechanism.
- Allows anyone to compete to become decision-making authority. Consensus mechanism governs this competition.
- applied in varying degrees from a semi-decentralized model to a fully decentralized one depending on the requirements and circumstances
- provides a way to remodel existing applications and paradigms, or to build new applications, to give full control to users

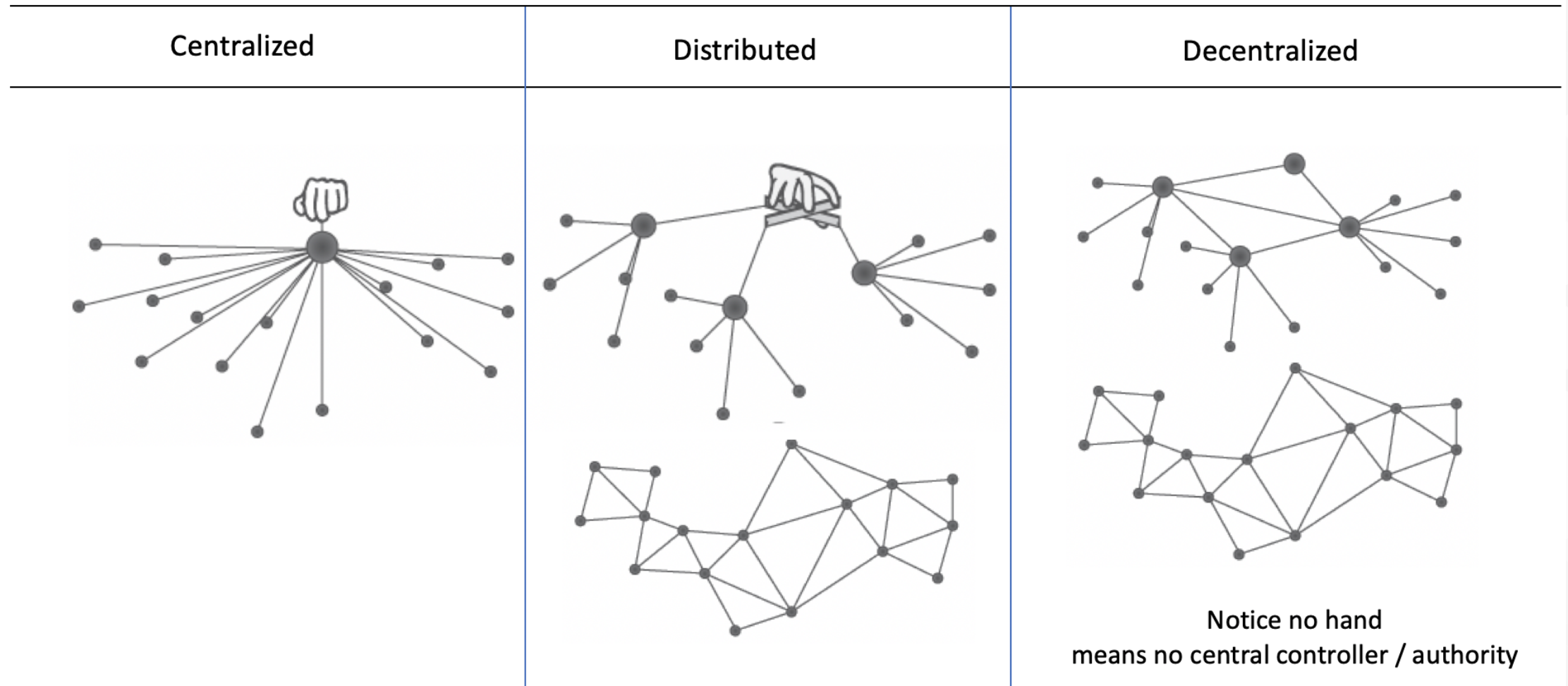
- **Information and communication technology (ICT)**
- centralized paradigm
- decentralized system and operate it with no single point of failure or single trusted authority
- run autonomously or by requiring some human intervention depending on the type and model of governance used in the decentralized application running on the blockchain

Defining decentralization



- **Centralized systems**
- **distributed systems**
 - data and computation are spread across multiple nodes in the network
 - parallel computing
- **decentralized system**
 - nodes are not dependent on a single master node
 - **decentralized consensus**
- The critical difference between a decentralized system and distributed system is that in a distributed system, there is still a central authority that governs the entire system, whereas in a decentralized system, no such authority exists.

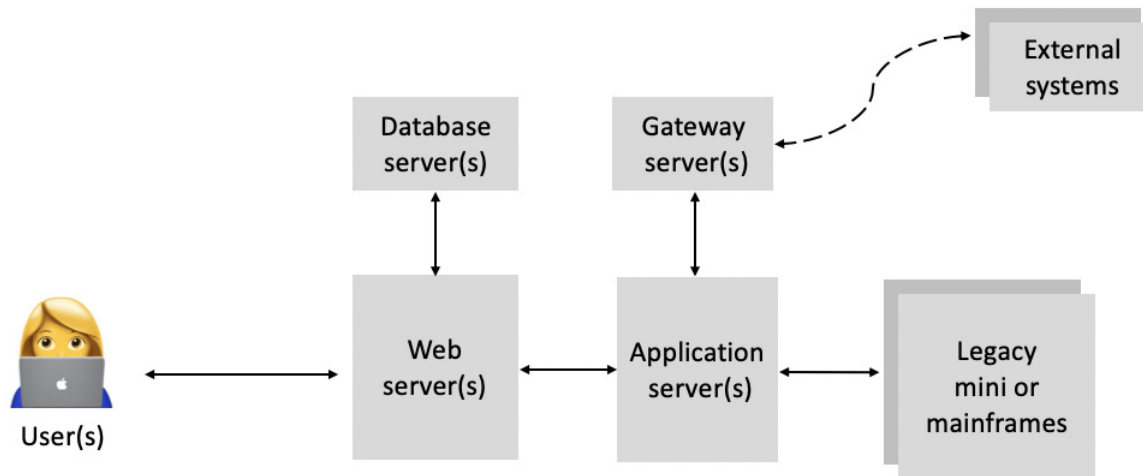
Decentralization using blockchain



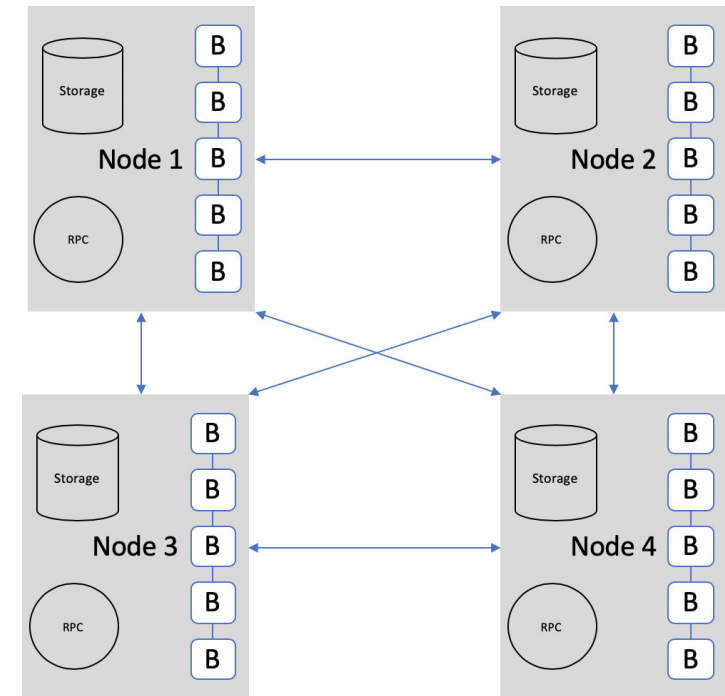
- The preceding diagram shows that the centralized model is the traditional one in which a central controller exists, and it can be viewed as a depiction of the usual client/server model.
- In the middle we have distributed systems, where we still have a central controller but the system comprises many dispersed nodes.
- On the right-hand side, notice that there is no hand/controller controlling the networks.
- This is the **key difference between decentralized and distributed** networks. A decentralized system may look like a distributed system from a topological point of view, but it doesn't have a central authority that controls the network.

Differences between distributed and decentralized systems

Distributed



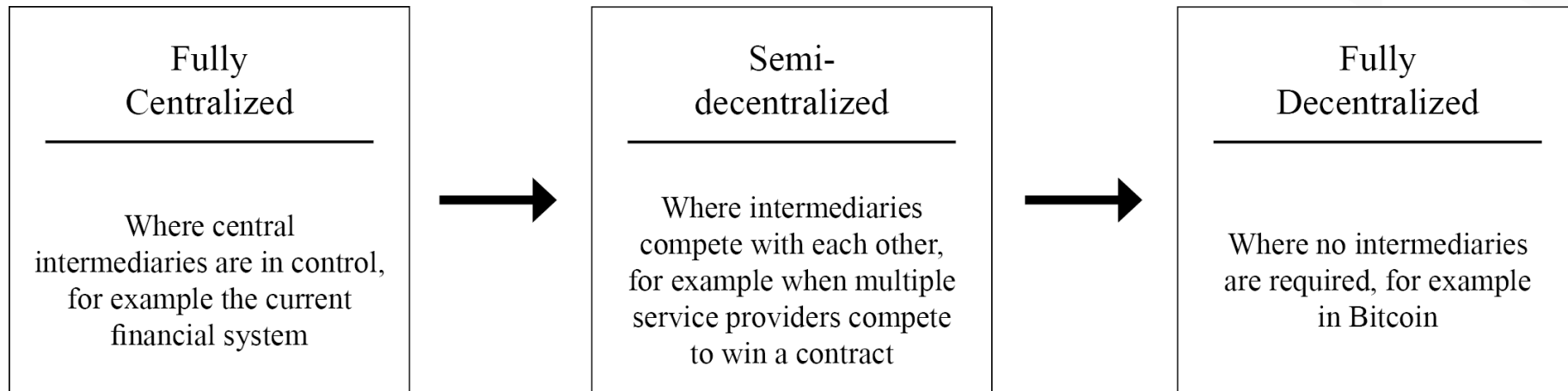
Decentralized



Feature	Centralized	Decentralized
Ownership	Service provider	All users
Architecture	Client/server	Distributed, different topologies
Security	Basic	More secure
High availability	No	Yes
Fault tolerance	Basic, single point of failure	Highly tolerant, as service is replicated
Collusion resistance	Basic, because it's under the control of a group or even single individual	Highly resistant, as consensus algorithms ensure defense against adversaries
Application architecture	Single application	Application replicated across all nodes on the network
Trust	Consumers have to trust the service provider	No mutual trust required
Cost for consumer	Higher	Lower

Methods of decentralization

- Disintermediation
- Contest-driven decentralization



- There are many benefits of decentralization, including transparency, efficiency, cost saving, development of trusted ecosystems, and in some cases privacy and anonymity.
- Some challenges, such as security requirements, software bugs, and human error, need to be examined thoroughly.
- For example, in a decentralized system such as Bitcoin or Ethereum where security is normally provided by private keys, how can we ensure that an asset or a token associated with these- private keys cannot be rendered useless due to negligence or bugs in the code? What if the private keys are lost due to user negligence? What if due to a bug in the smart contract code the decentralized application becomes vulnerable to attack?
- **understand that not everything can or needs to be decentralized**

Question	Yes/No	Recommended solution
Is high data throughput required?	Yes	Use a traditional database.
	No	A central database might still be useful if other requirements are met. For example, if users trust each other, then perhaps there is no need for a blockchain. However, if they don't or trust cannot be established for any reason, blockchain can be helpful.
Are updates centrally controlled?	Yes	Use a traditional database.
	No	You may investigate how a public/private blockchain can help.
Do users trust each other?	Yes	Use a traditional database.
	No	Use a public blockchain.
Are users anonymous?	Yes	Use a public blockchain.
	No	Use a private blockchain.
Is consensus required to be maintained within a consortium?	Yes	Use a private blockchain.
	No	Use a public blockchain.
Is strict data immutability required?	Yes	Use a blockchain.
	No	Use a central/traditional database.

- Answering all of these questions can help you decide whether or not a blockchain is required or suitable for solving the problem.
- Beyond the questions posed in this model, there are many other issues to consider, such as latency, choice of consensus mechanisms, whether consensus is required or not, and where consensus is going to be achieved. If consensus is maintained internally by a consortium, then a private blockchain should be used; otherwise, if consensus is required publicly among multiple entities, then a public blockchain solution should be considered. Other aspects, such as immutability, should also be considered when deciding whether to use a blockchain or a traditional database. If strict data immutability is required, then a public blockchain should be used; otherwise, a central database may be an option.

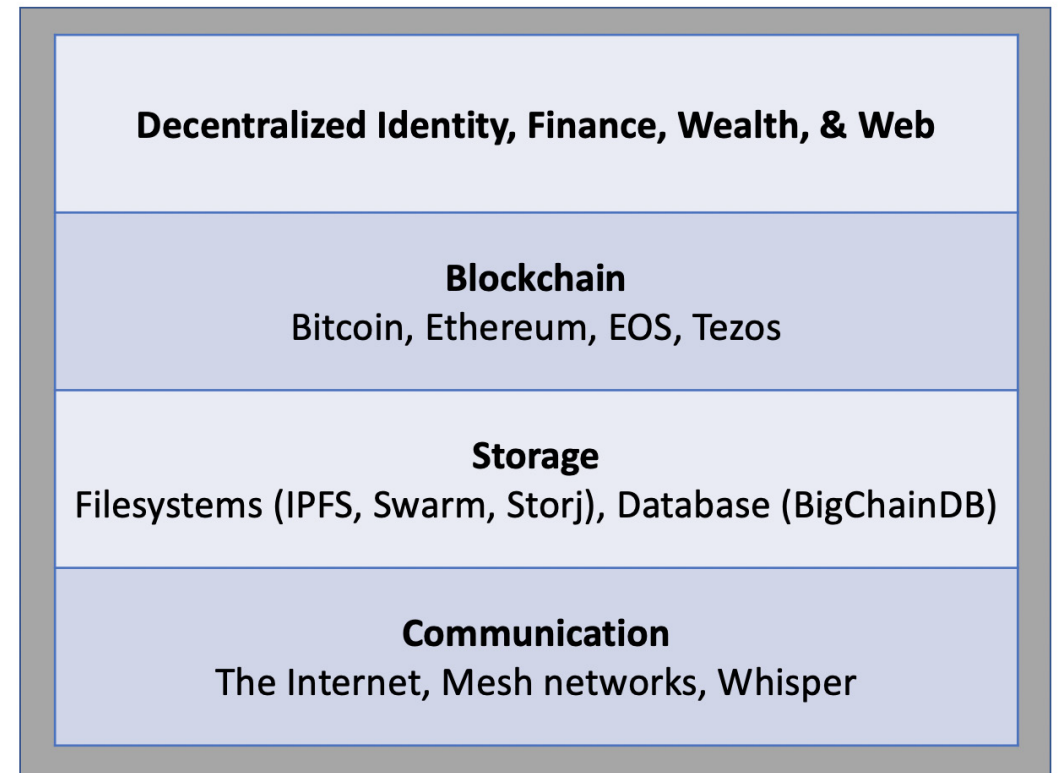
How to decentralize

- What is being decentralized?
- identify what system is being decentralized. This can be any system, such as an identity system or a trading system
- What level of decentralization is required?
- specify the level of decentralization required by examining the scale of decentralization. It can be full disintermediation or partial disintermediation.
- What blockchain is used?
- determine which blockchain is suitable for a particular application
- What security mechanism is used?
- how the security of a decentralized system will be guaranteed

Blockchain and full ecosystem decentralization

This model illustrates how a full decentralized ecosystem could work. It includes:

- Communication
- Storage
- Computing power
- Identity and wealth



Pertinent terminology

Smart contracts

- A **smart contract** is a software program that usually runs on a blockchain.
- Smart contracts do not necessarily need a blockchain to run; however, due to the security benefits that blockchain technology provides, blockchain has become a standard decentralized execution platform for smart contracts.
- A smart contract usually contains some business logic and a limited amount of data.
- The business logic is executed if specific criteria are met.
- Actors or participants in the blockchain use these smart contracts, or they run autonomously on behalf of the network participants.

Autonomous agents

- An **Autonomous Agent (AA)** is an artificially intelligent software entity that acts on the behalf of its owner to achieve some desirable goals without requiring any or minimal intervention from its owner.

Decentralized organizations

- DOs are software programs that run on a blockchain and are based on the idea of actual organizations with people and protocols.
- Once a DO is added to the blockchain in the form of a smart contract or a set of smart contracts, it becomes decentralized and parties interact with each other based on the code defined within the DO software

Decentralized autonomous organizations

- Just like DOs, a **decentralized autonomous organization (DAO)** is also a computer program that runs on top of a blockchain, and embedded within it are governance and business logic rules.
- DAOs and DOs are fundamentally the same thing. The main difference, however, is that DAOs are autonomous, which means that they are fully automated and contain artificially intelligent logic. DOs, on the other hand, lack this feature and rely on human input to execute business logic.
- In a DAO, the code is considered the governing entity rather than people or paper contracts. However, a human curator maintains this code and acts as a proposal evaluator for the community.

Decentralized autonomous corporations

- **Decentralized autonomous corporations (DACs)** are similar to DAOs in concept, though considered to be a subset of them.
- The definitions of DACs and DAOs may sometimes overlap, but the general distinction is that DAOs are usually considered to be nonprofit, whereas DACs can earn a profit via shares offered to the participants and to whom they can pay dividends.
- DACs can run a business automatically without human intervention based on the logic programmed into them.

Decentralized autonomous societies

- **Decentralized autonomous societies (DASes)** are a concept whereby an entire society can function on a blockchain with the help of multiple, complex smart contracts and a combination of DAOs and **decentralized applications (DApps)** running autonomously.
- This model does not necessarily translate to a free-for-all approach, nor is it based on an entirely libertarian ideology; instead, many services that a government commonly offers can be delivered via blockchains, such as government identity card systems, passports, and records of deeds, marriages, and births.

Decentralized applications

- DAOs, DACs, and DOs are DApps that run on top of a blockchain in a peer-to-peer network. They represent the latest advancement in decentralization technology.
- Decentralized Applications (DApps) can be categorized into Type 1 ,Type 2 or Type 3.
- **Type 1:** Run on their own dedicated blockchain, for example, standard smart contract based DApps running on Ethereum. If required, they make use of a native token, for example, ETH on Ethereum blockchain.
- For example, Ethlance is a DApp that makes use of ETH to provide a job market. More information about Ethlance can be found at <https://ethlance.com>.

- **Type 2:** Use an existing established blockchain. that is, make use of Type 1 blockchain and bear custom protocols and tokens, for example, smart contract based tokenization DApps running Ethereum blockchain.
- An example is DAI, which is built on top of Ethereum blockchain, but contains its own stable coins and mechanism of distribution and control. <https://makerdao.com/en/>
- Another example is Golem, which has its own token GNT and a transaction framework built on top of Ethereum blockchain to provide a **decentralized marketplace** for computing power where users share their computing power with each other in a peer-to-peer network. <https://golem.network>
- OMNI network, is a software layer built on top of Bitcoin to support trading of custom digital assets and digital currencies. <https://www.omnilayer.org>

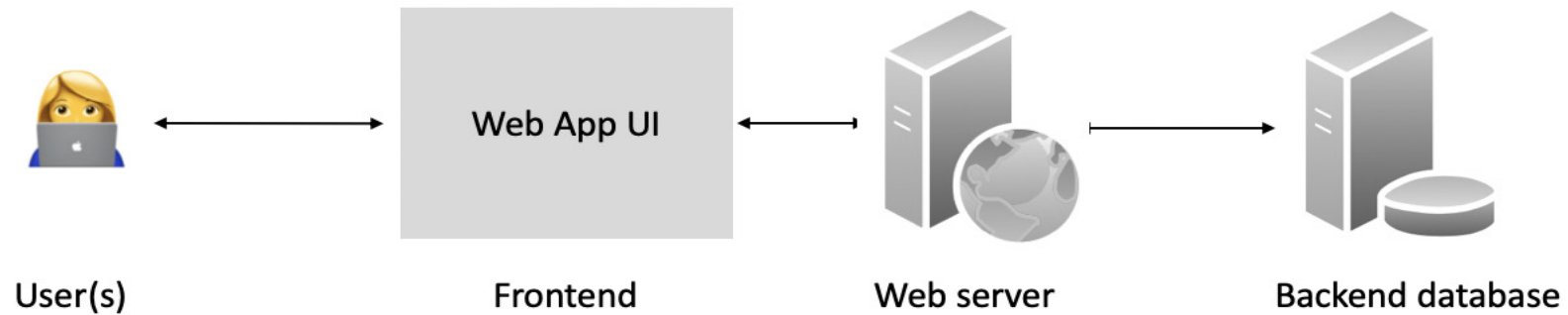
- **Type 3:** Use the protocols of Type 2 DApps; for example, the SAFE Network uses the OMNI network protocol. <https://safenetwork.tech>
- In the last few years, the expression DApp has been increasingly used to refer to any end-to-end decentralized blockchain application, including a user interface (usually a web interface), smart contract(s), and the host blockchain.
- There are thousands of different DApps running on various platforms (blockchains) now. There are various categories of these DApps covering media, social, finance, games, insurance, and health. There are various decentralized platforms (or blockchains) running, such as Ethereum, EOS, NEO, Loom, and Steem.
- The highest number of DApps currently is on Ethereum.

Requirements of a DApp

- The DApp should be fully open source and autonomous, and no single entity should be in control of a majority of its tokens. All changes to the application must be consensus driven based on the feedback given by the community.
- Data and records of operations of the application must be cryptographically secured and stored on a public, decentralized blockchain to avoid any central points of failure.
- A cryptographic token must be used by the application to provide access for and incentivize those who contribute value to the applications, for example, miners in Bitcoin.
- The tokens (if applicable) must be generated by the decentralized application using consensus and an applicable cryptographic algorithm. This generation of tokens acts as a proof of the value to contributors (for example, miners).

Design of a DApp

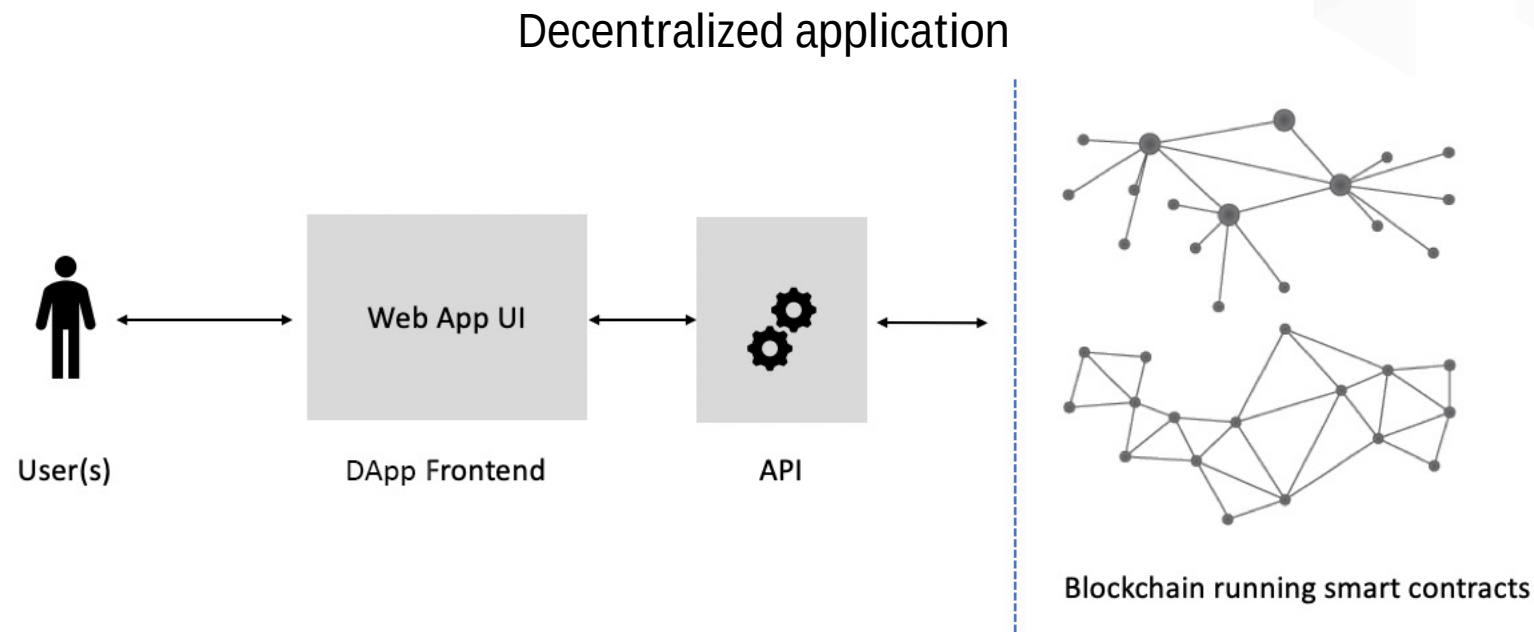
Traditional client server model



Traditional apps commonly consist of a user interface and usually a web server or an application server and a backend database. This is a common client/server architecture

Design of a DApp

- A DApp on the other hand has a blockchain as a backend and can be visualized as depicted in the following diagram. The key element that plays a vital role in the creation of a DApp is a smart contract that runs on the blockchain and has business logic embedded within it:



- Note that the frontend in either a DApp or app architecture can either be a thick client, a mobile app, or a web frontend (a web user interface). However, it is usually a web frontend commonly written using a JavaScript framework such as React or Angular.

DO, DAO, DAC, DAS, DApp

The table highlights some key properties of some different types of decentralized entities; Decentralized Organizations (DOs), Decentralized Autonomous Organizations (DAOs), Decentralized Autonomous Corporations (DACs), Decentralized Autonomous Societies (DASes), and DApps

Entity	Autonomous?	Software?	Owned?	Capital?	Legal status	Cost
DO	No	No	Yes	Yes	Yes	High
DAO	Yes	Yes	No	Yes	Unsettled	Low
DAC	Yes	Yes	Yes	Yes	Unsettled	Low
DAS	Yes	Yes	No	Possible	Unsettled	Low
DApp	Yes	Yes	Yes	Optional tokens	Unsettled	Use case dependent

Platforms for decentralization

- Ethereum
- MaidSafe
- Lisk
- EOS

Innovative trends

- Decentralized web
 - Web 1: the original World Wide Web
 - Web 2: the era when more service-oriented and web-hosted applications started to emerge
 - Web 3: the vision of the decentralized internet or web
- Decentralized identity
- Decentralized finance (DeFi)

Summary

In this chapter we:

- Defined the concept of decentralization.
- Introduced DApps, applications based on a decentralized architecture.
- Considered decentralization from the blockchain perspective.
- Looked at some key features of DAOs, DACs, and Dapps, among other decentralized entities.
- Explored some innovative trends.